

DEXTER FERNANDES

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SUMMARY

Backend engineer with 3+ years building and operating Python microservices in event-driven, cloud-native architectures, from Kafka-based reliability patterns on Dockerised inference services to containerised REST APIs on Microsoft Azure. Comfortable owning a service end to end: design, tests, deployment, monitoring and production support. Cut live pipeline crashes by 67% through health checks, retries and structured observability, and drove test-first development across service and inference code. MSc Robotics, University of Bristol.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Hardened deployed Python services with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Worked test-first on the Python inference services, writing failing pytest cases against expected detection and health-check contracts before implementing them, which kept interfaces stable as the pipeline changed underneath.
- Implemented reliability patterns across the service layer using timeouts, retries and message validation, with structured logs and metrics for root cause analysis under load.
- Introduced pytest, unittest and linting into CI for edge inference modules, improving release consistency across hardware targets.
- Wrote regression tests reproducing each production failure before fixing it, turning field incidents into a permanent test suite and stopping the same crashes recurring across deployments.
- Developed a watchdog application that detected and restarted failed services on edge devices autonomously, removing the need for engineer callouts.
- Owned production monitoring and post-deployment validation for deployed services, using observability tooling and drift tracking to catch issues before they reached operators.
- Diagnosed and resolved production failures through structured analysis and validation checklists, and documented recovery procedures and deployment guardrails for the engineers supporting the fielded services.

LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Deployed containerised ML inference microservices on Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost per inference.
- Designed the PostgreSQL schema routing real-time incidents into downstream analytics, writing the joins and CTEs behind operational reporting.
- Established test-first practice across a four-engineer team, requiring a failing test alongside each ticket before implementation and reviewing test coverage as part of code review.
- Drove the C++ inference path test-first, specifying pre-processing, inference and post-processing behaviour as gtest cases before implementation, which made an async refactor from 5 to 15 FPS safe to attempt.
- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency and operational cost while holding safety-critical performance targets.
- Mentored junior engineers and worked with hardware, software and QA teams to integrate perception modules into deployed systems.
- Led four engineers building a two-stage detector for a safety-classification workflow, setting experimentation standards and reviewing model performance, and cutting false positives by 33% within latency and power budgets.
- Designed data handling and retention practices aligned with GDPR principles, including data minimisation and controlled access.

PROJECTS

DeepStream RTSP Video Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline handling video ingest, detection and re-streamed output across 3 concurrent feeds, with metadata extraction and anonymisation.
- Converted a detection model to a dynamic-batch TensorRT FP16 engine on constrained VRAM, validating accuracy against FP32 and profiling latency to sustain real-time throughput per stream.
- Implemented a C++ and CUDA TensorRT plugin to move post-processing into the inference engine, and benchmarked three object trackers on accuracy and resource use to guide selection.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/Modern-Edge-CV-Benchmarking

03/2026

- Built a repeatable Python benchmarking harness comparing eight detection and segmentation model families on accuracy, latency, model size and deployment suitability.
- Exported models to ONNX and evaluated OpenVINO deployment paths against edge constraints, producing config-driven, reproducible runs.
- Produced structured comparison reports enabling regression checks between benchmark runs.

FinanceBench RAG Evaluation Harness

06/2026

- Built a Python evaluation framework measuring how well RAG and document-grounded QA systems answer using the correct source evidence, with LLM-as-judge scoring, evidence matching and hallucination labels.
- Designed structured experiments across closed-book, oracle-context and dense RAG workflows to isolate whether failures came from retrieval, source selection or generation.
- Produced reproducible runs with automated tests, config-driven experiments and structured outputs, enabling regression checks between runs.

ROS2 SLAM and Nav2 Simulation

02/2026

- Developed ROS2 nodes, launch files and parameterised configurations in C++ and Python, using message synchronisation and timestamp alignment to debug transform timing issues.
- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, plus an EKF localisation layer fusing wheel odometry and IMU.
- Containerised the stack with Docker and managed versioning in Git for reproducible builds.

EDUCATION

University of Bristol — MSc Robotics, Merit 09/2023 – 09/2024 · BRISTOL, UK
Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time system in C++17 and ROS2 on a Clearpath Husky, integrating LiDAR, depth and radiological sensing for inspection mapping.
- Benchmarked three SLAM solver backends under a single evaluation protocol across matched sensor sequences.
- Configured and tuned Ceres, g2o and GTSAM solver backends and ICP parameters, resolving transform timing issues across a point cloud processing pipeline.

Visvesvaraya National Institute of Technology, Nagpur — BTech Electrical and Electronics Engineering 07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in electrical and electronics engineering, mathematics and systems fundamentals underpinning subsequent software work.

SKILLS

Languages	Python, C++17, SQL, Bash
Backend and cloud	FastAPI, Django, REST API design, microservices architecture, Docker, Apache Kafka, event-driven architecture, PostgreSQL, Microsoft Azure, queue-based async processing, autoscaling
Testing and CI/CD	test-driven development (TDD), pytest, unittest, gtest, CI/CD pipelines, automated testing, linting
Operations	structured logging, observability, health checks, timeouts and retries, production monitoring, regression testing, root cause analysis, Linux, Git
Video pipelines	NVIDIA DeepStream, GStreamer, RTSP video ingest, TensorRT, metadata extraction, latency profiling
ML and robotics tooling	PyTorch, ONNX, YOLO, ROS2 (rclcpp, rclpy), PCL, Open3D, GTSAM, Ceres, g2o, RAG pipelines, LangChain

ADDITIONAL EXPERIENCE

GrowthStage — Technical Screening Engineer (Contract) 05/2026 – PRESENT · REMOTE
Assess candidate technical depth across perception, SLAM and deployment engineering, running hands-on technical screens and writing and calibrating evaluation rubrics.

Independent Engineering — Self-Directed Projects 09/2024 – 04/2026 · BRISTOL, UK
Self-directed engineering programme covering detection architectures, edge deployment, retrieval systems and evaluation tooling, delivered as eight open-source projects with published benchmarks, including the DeepStream RTSP pipeline and RAG evaluation harness detailed under Projects.

CERTIFICATIONS AND AWARDS

Building RAG Agents with LLMs, NVIDIA (06/2026) Deep Learning Specialisation, deeplearning.ai (07/2019)